Taking log of a matrix equation

Asked 4 years, 7 months ago Modified 4 years, 7 months ago Viewed 2k times

1

I have the following relation

$$\rho = \frac{1}{x}\exp(\log(\rho) + \delta),$$

where x is a number and ρ and δ are matrices of size $n \times n$. I would like to solve this for δ. I take logs on both sides and get

$$\log \rho = \log \rho + \delta - \log x$$

What I have written here is dimensionally inconsistent since x is a number and the other terms are matrices. But I'm not sure what's wrong since taking the log seems okay. Also it seems okay to use $\log(A/B) = \log(A) - \log(B)$ since x is just a number and commutes with all the matrices.

What exactly is the right solution for δ? Is it $\delta = xI$, where I is the appropriately sized identity matrix? Or are there some technicalities that I'm missing?

linear-algebra logarithms

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edited May 27, 2019 at 22:20

asked May 27, 2019 at 22:14
user1936752
1,656 1 14 28

3 you can just replace $\frac{1}{x}$ with $\frac{1}{x}I$ and now everything in your equation is a matrix. – Set May 27, 2019 at 22:26

2 Answers

Sorted by: Highest score (default)

3

It's true that, under some assumptions, $\log(ab) = \log a + \log b$ for scalars ("numbers") and $\log(AB) = \log A + \log B$ for commuting matrices. But it's obviously false that $\log(aB) = \log a + \log B$.

Hence, you must be careful. What you can write is

$$\log aB = \log aIB = \log(aI) + \log(B)$$

Further, consider $e^{tI} = \sum_n t^n I^n/n! = e^t I$, hence $tI = \log(e^t I) \implies \log(aI) = \log(a)I$.

Then

$$\log \frac{1}{x}M = -\log(x)I + \log(M)$$

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answered May 27, 2019 at 23:54
leonbloy
62.6k 10 74 153

1

Your work looks correct, and it is perhaps the case (as it was in my math classes) that your notation is the standard one; but it is confusing. Notice that

$$\frac{1}{x} = \frac{1}{x}I$$

where I is the $n \times n$ identity matrix. You can then distribute that x^{-1} into the identity matrix to get a diagonal matrix with x^{-1} as all of its elements. It becomes clear then that scalar multiples can always to be distributed into an identity matrix.

What you take away from this is that x^{-1} is fine when we are working with multiplication, but once we perform operations like log that separate things into addition, x^{-1} becomes a matrix with no change of notation, it is just understood that it is now a matrix. Which, of course, is confusing.

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answered May 27, 2019 at 22:50

Kraigolas
1,549 5 19

The equation is false, as you have an escalar on the left and a matrix on the right. – leonbloy May 31, 2019 at 3:27

1 @leonbloy it's not an equality, it's a definition. I've seen the I omitted often as it is just *understood* that this is what it means. – Kraigolas May 31, 2019 at 10:37